SIMULATION EXPERIMENT – 2

❖ Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim:

To simulate and analyze the performance of microwave transmission lines and two-port networks over a specified frequency range.

Objectives of this Experiment:

1. To model coaxial, waveguide or microstrip transmission structures for microwave-frequency operation.

2. To determine characteristic impedance, propagation constant and transmission characteristics of the selected structure.

3. To obtain and analyze S11, S21, S12 and S22 over a specified frequency range.

4. To evaluate return loss, insertion loss and transmission efficiency from the simulated S-parameter response.

Software Required:MATLAB / ADS / CST / HFSS

Matlab Code:

```
clc; clear; close all;
f=8:0.05:12; f0=10; Z0=50; L=50e-3;

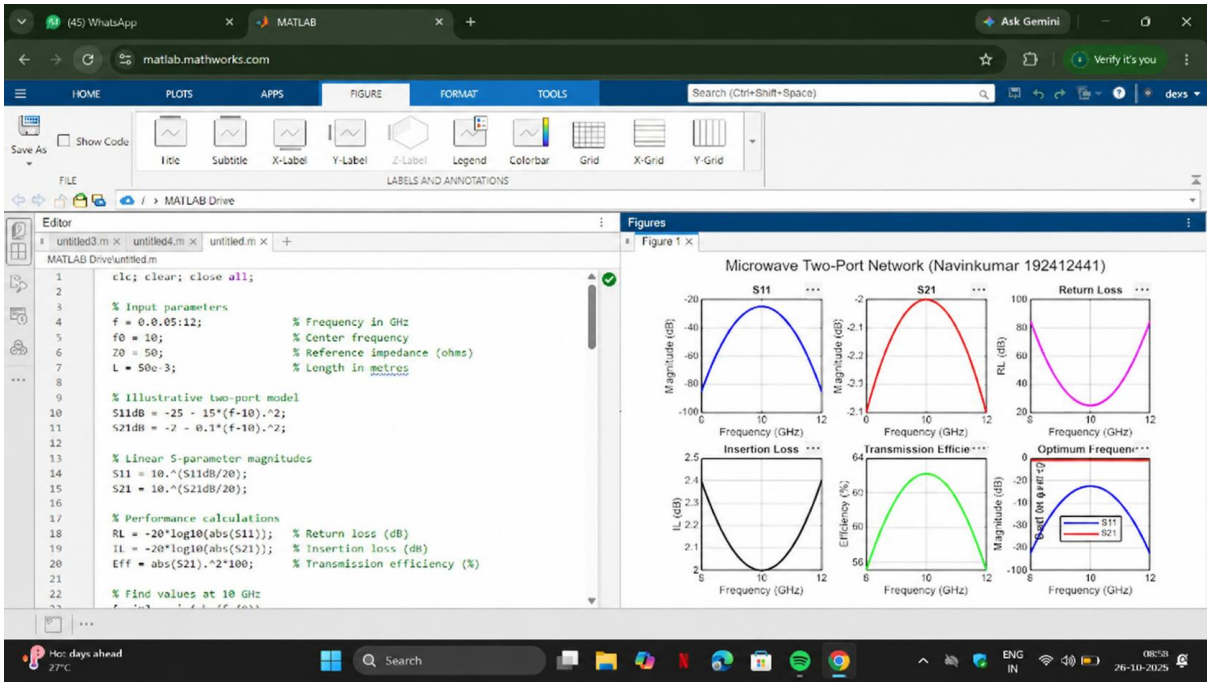
S11dB=-25-15*(f-f0).^2;
S21dB=-2-0.1*(f-f0).^2;
S11=10.^(S11dB/20); S21=10.^(S21dB/20);

RL=-20*log10(S11);
IL=-20*log10(S21);
Eff=S21.^2*100;

fprintf('S11=%.2f dB\n',S11dB(41));
fprintf('S21=%.2f dB\n',S21dB(41));
```

```
fprintf('Return Loss=%.2f dB\n',RL(41));
fprintf('Insertion Loss=%.2f dB\n',IL(41));
fprintf('Efficiency=%.2f%%\n',Eff(41));

figure;
subplot(2,3,1); plot(f,S11dB); title('S11'); grid on;
subplot(2,3,2); plot(f,S21dB); title('S21'); grid on;
subplot(2,3,3); plot(f,RL); title('Return Loss'); grid on;
subplot(2,3,4); plot(f,IL); title('Insertion Loss'); grid on;
subplot(2,3,5); plot(f,Eff); title('Efficiency'); grid on;
subplot(2,3,6); plot(f,S11dB,f,S21dB);
title('Optimum Frequency'); legend('S11','S21'); grid on;
```



Result:

Thus the S-parameters, return loss, insertion loss, and transmission efficiency were calculated and plotted over 8–12 GHz.The optimum operating frequency is 10 GHz, with S11 = −25 dB, S21 = −2 dB, and transmission efficiency ≈ 63.10%.

$\beta = 209.44$ rad/m

3. reflective co-efficient

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50-50}{50+50}$$

$$S_{11} = |\Gamma| = 0$$

$S_{21} = e^{-j\beta L}$

$|S_{21}| = 1$

$\beta = 209.44$

$L = 10.47$ rad (0.05)

$|S_{21}| = 1$

4. Efficiency

$$\eta = |S_{21}|^2 \times 100\%$$

$\eta = 100\%$

Return loss

$$R_L = -20 \log_{10} |S_{11}|$$

$R_L = \infty$ dB

Insertion loss

$$I_L = -20 \log_{10} |S_{21}|$$

$I_L = 0$ dB